

DE Lab3
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Q1: Answer the following questions for the circuit below:

a) Calculate V_{OH} , V_{OL} , V_{IL} , V_{IH} , and V_S .

From excel setting $V_{in} = V_{IL}$, calculating the LHS and RHS of the equation, and using goal seek to set the difference to 0 using V_{out} :

$V_{OH}=1.199453382$

$V_{OL}=0.001113032$

$V_{IL}=0.536212065$

$V_{IH}=0.53805131$

$V_{Chi}=1.414213562$

$V_S=0.565685425$

b) Describe in words the relative change in the part(a) voltages if the width of the PMOS transistor doubles.

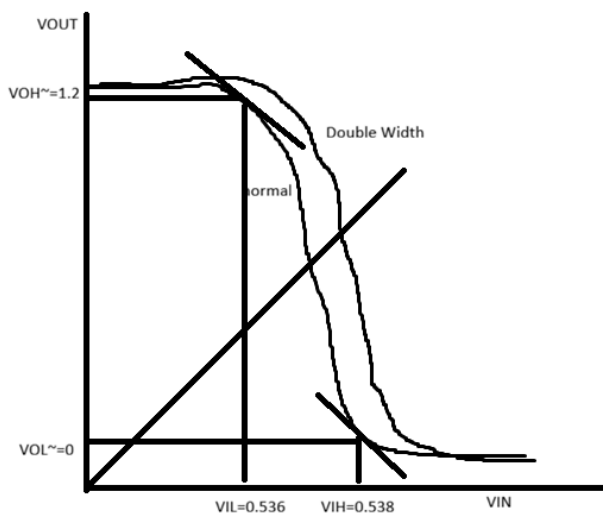
V_{IL} and V_{IH} are increased, V_{OH} and V_{OL} remain about the same, V_S increases

c) Calculate V_S again with a double width PMOS.

$V_{Chi}=1$

$V_S=0.6$

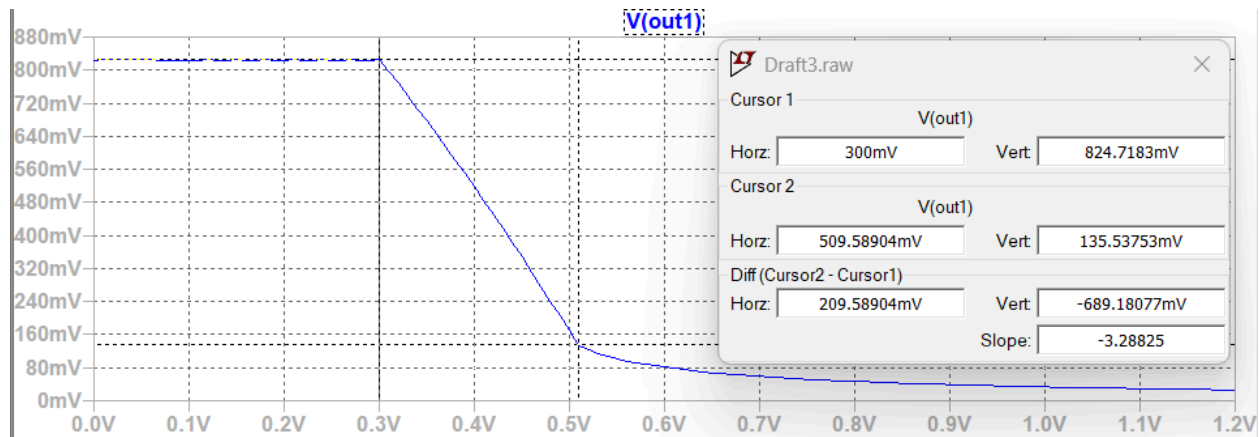
d) Sketch the VTC (Voltage Transfer Curve) for the inverter in both PMOS width cases in one plot.



Q2: Answer the following questions for the circuit below:

a. Using LTSpice determine the noise margins by plotting the VTC.

i. Remember to set capacitor initial conditions if needed.



NMlow=0.3V

NMhigh=1.2V-0.509V=0.691V

b. Calculate V_S by hand and hand draw the VTC. Do not bother with V_{IL} , V_{IH} calculation.

i. How does this compare to the simulated VTC?

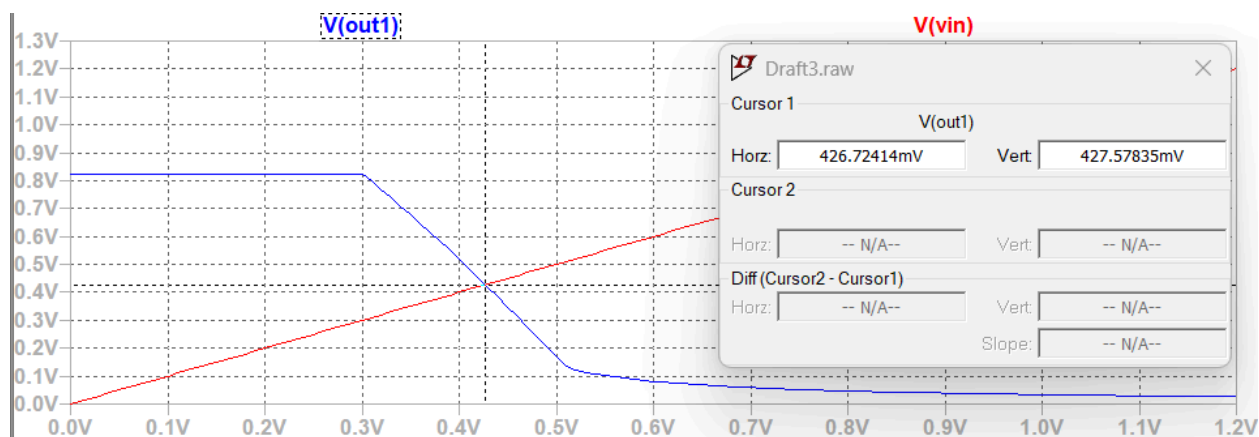
Calculated:

$$X = \sqrt{((1\mu/0.6)/(0.1\mu/0.6))} = \sqrt{10} = 3.16$$

$$V_S = (V_{DD} - |V_{TN}| + X \cdot V_{TN}) / (1 + X) = (1.2 - 0.4 + 3.16 \cdot 0.4) / (1 + 3.16) = 0.496V$$

$V_S = 0.496V$

Simulated



$V_S = 0.427$

Given we hand calculated by approximation, these values are very close.